

ARYAN SETHI

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EDUCATION

National University of Singapore <i>M.Sc. Financial Engineering (Part-Time)</i>	Aug 2026 - May 2028
Nanyang Technological University, Singapore <i>B.Eng. Computer Science, Specialization in Artificial Intelligence</i> Honours (Distinction), CGPA: 4.3 / 5.0	Aug 2021 - May 2025

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, SQL, Kotlin, Dart, HTML/CSS
Backend: .NET Core, .NET WebForms, SpringBoot, Golang, Node.js, Microservices, REST APIs, gRPC, RabbitMQ
Frontend & Mobile: HTML, CSS, JavaScript, React, Vue, Angular, Flutter
Databases: MySQL, SQL Server, PostgreSQL, MongoDB, OpenSearch, Redis
ML & Analytics: Deep Reinforcement Learning, PyTorch, LangChain, LangGraph, RAG, Scikit-Learn, Pandas, NumPy
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Solace
Systems: Linux, Shell Scripting, Task Scheduling, Multithreading, System Architecture

EXPERIENCE

MRI Software <i>Software Engineer 1</i> - Built a reconciliation job comparing computed gross-turnover rent against posted POS sales feeds before invoice release, catching ~30 mis-billed tenant invoices per cycle that previously surfaced as credit notes. - Built a validation layer over arrears interest computation that flags allocations inconsistent with lease payment terms, cutting monthly interest reversals from ~20 to under 5. - Built lease and license space management modules that helps clients create an audit trail capturing every change to spaces set up for leasing/licensing, reducing dispute investigations for client teams. - Refactored core tenancy modules within a large legacy .NET Web Forms codebase to reduce coupling and improve readability, cutting the time required to extend affected areas by 60%. - Built automation tool that converts website projects into web application projects, enabling static analysis across repositories and reducing manual setup from 3 hours to 5 minutes per project. - Gathered requirements directly from enterprise clients and validated feature behaviour before release, reducing post-release change requests and improving user experience. <i>Tech: .NET Web Forms, C#, React, HTML, CSS, SQL Server, Python, Azure, GitLab CI/CD, REST APIs, gRPC</i>	Aug 2025 - Present
GetGo Carsharing <i>Software Engineering Intern</i> - Designed and deployed a feature flag service across multi-environment microservices, enabling feature toggles without redeployment and accelerating deployment cycles by 30% while reducing rollback incidents by 25%. - Published an internal private package library for the feature flag service, adopted by 100% of internal services and reducing integration effort. - Engineered fallback mechanisms on the Solace event broker with session recovery and automated failover, raising system availability from 99.2% to 99.9%. - Developed a company-wide DevOps Management System dashboard centralising infrastructure monitoring and operations, improving incident response time by 35% across 5 engineering teams. <i>Tech: .NET Core, C#, Golang, React, Python, HTML, CSS, MySQL, OpenSearch, Redis, AWS, Solace, Docker, Kubernetes, Microservices, REST APIs, gRPC</i>	Dec 2024 - Apr 2025
GetGo Carsharing <i>Junior Software Developer Intern</i> Built a high performance search service to absorb heavy-load query traffic, reducing infrastructure costs by 10% and improving p95 search latency.	Jan 2024 - Aug 2024

- Created a mock service replicating third-party vendor responses, unblocking performance testing for scenarios that could not be reproduced in live environments and shortening integration timelines.
- Developed a data synchronisation tool to reconcile records between MySQL and OpenSearch, eliminating recurring production inconsistencies and cutting data-integrity incidents.
- Implemented an aggregator service consolidating responses across microservices, reducing inter-service calls per request and simplifying frontend integration.

Tech: .NET Core, C#, Golang, React, Python, HTML, CSS, MySQL, OpenSearch, Redis, AWS, Solace, Microservices, REST APIs, gRPC

PROJECTS

NTUSwap, Decentralized Exchange Platform

Sep 2024 - Dec 2024

- Built a decentralised exchange in React and Solidity for peer-to-peer ERC-20 token trading, processing over \$10K in test volume with zero failed transactions.
- Developed smart contracts for token swaps and liquidity provision using OpenZeppelin, reaching 95% test coverage.
- Integrated Web3.js and MetaMask and deployed to Ethereum testnets for end-to-end validation.

Project Repository - github.com/RowenTey/ntuswap

Sentiment Classification, Natural Language Processing

Sep 2024 - Dec 2024

- Built and compared RNN, biLSTM, biGRU, and CNN architectures for sentiment analysis on Rotten Tomatoes reviews, reaching 79% baseline accuracy with improved out-of-vocabulary handling.
- Fine-tuned DistilBERT to 90.62% and packaged the model as a RESTful microservice, deployed to AWS SageMaker for inference, reducing manual labelling/processing time by 40%.
- Developed an NLP preprocessing pipeline covering hyphenation, named entity recognition, and non-English translation.

Project Repository - github.com/AryanSethi20/sentiment-classification

PostgreSQL Query Explorer - Database System Design and Principles

Oct 2023 - Dec 2023

- Developed a visualisation tool for PostgreSQL execution plans with real-time block access tracking and heatmaps, covering 95% of analysed queries.
- Built a Flask backend integrated with PostgreSQL to process TPC-H datasets and surface cost and operator-node insights.
- Implemented block-level tracking and query diagnostics to expose optimisation opportunities across diverse SQL workloads.

Project Repository - github.com/AryanSethi20/Database-System-Principles-Project

Database Design - C++ Storage Engine with B+ Tree Index

Aug 2023 - Sep 2023

- Engineered a C++ database system for NBA game data using compact record structures and unspanned storage, reducing storage footprint and disk I/O by 40%.
- Implemented a custom B+ Tree index supporting bulk loading, range search, and delete operations, cutting average query time by 70% against a linear-scan baseline.

Project Repository - github.com/indicium15/sc3020-project-1

PUBLICATIONS & RESEARCH

AI-Based Edge Caching for 5G-and-Beyond Industrial IoT Networks, Final Year Project

Aug 2024 - May 2025

- Implemented deep reinforcement learning policy switching with DQN and DDQN to minimise Peak Age of Information in 5G-enabled IIoT networks, improving on static strategies by over 10%.
- Built an MQTT-based edge caching system that toggles between Zero-Wait and Continuous Update policies in real time based on live network metrics.
- Benchmarked agent performance under varying traffic loads, demonstrating improved stability and decision quality from DDQN over standard DQN.

Project Repository - github.com/AryanSethi20/AI-Based-Edge-Caching-on-5G-and-Beyond-Industrial-IOT-Networks

ADDITIONAL

SAP S.C.A.L.E. 2023, First Runner-Up

Jun 2023 - Jul 2023

Delivered a digital transformation strategy for a beverage company using SAP cloud technologies, placing 2nd overall.

MIT \$15K Competition, Semi-Finalist

Jun 2022 - Sep 2022

Led "The Ambulant," an affordable canine wheelchair with a suspension system and mobility-tracking navigation bar, advancing to the semi-final round.